Yijie Xu 徐亦捷

Ph.D. Candidate in Artificial Intelligence at HKUST(GZ)
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RESEARCH INTEREST

His current research centers on how language models acquire and leverage information to adapt and make decisions, with a focus on **test-time self-evolution**, **generative retrieval**, and **retrieval-augmented generation (RAG)**.

EDUCATION

Hong Kong University of Science and Technology (Guangzhou) — *Guangzhou, China*
Ph.D. Candidate in Artificial Intelligence — *Aug. 2027 Expected*

- **Supervisor**: Prof. Hui Xiong, Associate Vice President, Founding Head of AI Thrust, Fellow of the ACM, IEEE, AAAS, AAAI, and CAAI.
- **Co-supervisor**: Prof. Xuming Hu, Assistant Professor in AI Thrust, HKUST(GZ).

Hong Kong University of Science and Technology (Guangzhou) — *Guangzhou, China*
M.Phil. in Artificial Intelligence — *Sep. 2022 - Aug. 2024*

- **CGPA**: 3.96/4.3
- **Supervisor**: Prof. Hui Xiong, Associate Vice President, Founding Head of AI Thrust, Fellow of the ACM, IEEE, AAAS, AAAI, and CAAI.
- **Co-supervisor**: Prof. Xiaojuan Ma, Associate Professor in CSE, HKUST.

University of Reading &
Nanjing University of Information Science and Technology — *Reading, England / Nanjing, China*
Dual B.Sc. in Mathematics and Applied Mathematics — *Sep. 2018 - Jun. 2022*

- **Dual-degree** program.
- **CGPA**: 3.71/4
- **Final Year Project**: Research on Interpolation and Data Fitting: Basis and Applications.

PUBLICATIONS

†: Co-first Authors; ‡: Co-second Authors; *: Corresponding Authors.

Retrieval & Retrieval-Augmented Generation

[1] **Generate, Then Match: Lightweight Relevance Modeling for Generative Retrieval**
Yijie Xu, Changhao Guan, Xingbo Yao, Chao Deng, Jie Liu, Jin Ma, Hui Xiong*
Preprint

[2] **SePer: Measure Retrieval Utility Through The Lens Of Semantic Perplexity Reduction**
Lu Dai, **Yijie Xu**, Jinhui Ye, Hao Liu*, Hui Xiong*
ICLR'25 Spotlight(3.20%): *The Thirteenth International Conference on Learning Representations*

Test-Time Self-Evolution & Agents

[3] **Unsupervised Post-Training of Foundation Models: A Survey**
Yijie Xu, Qianyi Cai, Huizai Yao, Yili Wang, Tianfu Wang, Cehao Yang, Xingbo Yao, Zhiyu Guo, Aiwei Liu, Xuming Hu*, Weiyu Guo*, Hui Xiong*
EMNLP'26: *The 2026 Conference on Empirical Methods in Natural Language Processing*

[4] **Rethinking Test-Time Adaptation for LLMs as Decoupled Unsupervised Post-Training**
Yijie Xu, Yili Wang, Qianyi Cai, Huizai Yao, Hanqing Wang, Tianfu Wang, Aiwei Liu, Weiyu Guo, Xuming Hu*, Hui Xiong*
Preprint

[5] **Towards Budget-Efficient Socially Intelligent Agents**
Linxiao Gong†, **Yijie Xu**†, Tianfu Wang, Yin Wu, Yili Wang, Xingbo Yao, Huizai Yao, Xilin Xia, Haowen Yang, Hui Xiong*
Preprint

[6] **You only need 4 extra tokens: Synergistic Test-time Adaptation for LLMs**
Yijie Xu, Huizai Yao, Zhiyu Guo, Weiyu Guo, Pengteng Li, Aiwei Liu, Xuming Hu, Hui Xiong*
Under Review for TPAMI: *IEEE Transactions on Pattern Analysis and Machine Intelligence*
Honored to be cited by **Prof. Yann LeCun** and **Jitendra Malik** as a representative exploration of LLM test-time adaptation.

[7] **Gated Test-Time Scaling for GUI Grounding via Multi-Seed Zoom Search**
Huizai Yao, **Yijie Xu**, Tianfu Wang, Yili Wang, Yi Cui, Yunfan Lu, Fengjie Zhu, Xuemin Zhao, Hengyao Bao, Ying Sun*, Hui Xiong*
Preprint

[8] **AutoSoP: Automated Software Profiling for GUI Agents**
Tianfu Wang, Linxiao Gong, Huizai Yao, Yunfan Lu, **Yijie Xu**, Yili Wang, Qianyi Cai, Fengjie Zhu, Xuemin Zhao, Hengyao Bao, Ying Sun, Hui Xiong*
Preprint

[9] **A Survey of Agent Skills: Toward Procedural Infrastructure for LLM Agents**
Cehao Yang, Xiaojun Wu, Honghao Liu, Xueyuan Lin, Chengjin Xu, Xuhui Jiang, Yuanliang Sun, Wenjie Zhang, Zhichao Shi, **Yijie Xu**, Jia Li, Hui Xiong*, Jian Guo*
Preprints

LLM Security

[10] **Detector-Aligned Watermark Distillation for Large Language Models**
Yijie Xu, Junyan Zhang, Xuming Hu, Hui Xiong*
Preprint

[11] **Mark Your LLM: Detecting the Misuse of Open-Source Large Language Models via Watermarking**
Yijie Xu†, Aiwei Liu†, Xuming Hu*, Lijie Wen, Hui Xiong*
WMARK@ICLR2025: *The 1st Workshop on GenAI Watermarking, ICLR 2025*
Extended Version Under Review for TKDD: *ACM Transactions on Knowledge Discovery from Data*

[12] **Machine Unlearning in Diffusion LLMs**
Yili Wang, **Yijie Xu**, Lu Dai, Tairan Huang, Qianyi Cai, Huizai Yao, Tianfu Wang, Hui Xiong*
NeurIPS'26 Poster: *The Fortieth Annual Conference on Neural Information Processing Systems*

[13] **VL-Eraser: Vacuum Distillation for Machine Unlearning in Vision-Language Models**
Yili Wang, Lu Dai‡, Tairan Huang‡, **Yijie Xu**‡, Hui Xiong*
CVPR'26: *The IEEE / CVF Computer Vision and Pattern Recognition Conference*

Data Mining

[14] **Against Political Polarization: A Unified Framework for Tracing Evolving Political Ideologies on Social Media**
Yijie Xu, Chao Wang*, Hui Xiong*
TIST: *ACM Transactions on Intelligent Systems and Technology*

[15] **SafeBuild-Bench: A Temporal-Robust Construction Safety Benchmark with Graph-Enhanced Data Mining**
Yi Cui†, Zilin Wang†, **Yijie Xu**, Qianyi Cai, Huizai Yao, Shuai Jiang, Bingzhuo Zhong*, Hui Xiong*
KDD Datasets & Benchmarks Track'26

[16] **Contrastive Graph Matching for Domain Adaptive Object Detection**
Pengteng Li, Huizai Yao, Yi Cui, Yunfan Lu, **Yijie Xu**, Minghao Yang, Pinhao Song
arXiv

Multimodal Learning & Perception

[17] **Two Drifts, One Principle: Conflict-Aware Spectral Consolidation for Multimodal Continual Learning**
Haiyi Zhang, Qianyi Cai, Hanqing Wang, Zilin Wang, Xinyue Di, **Yijie Xu**, Yili Wang, Tianfu Wang, Yifan Han, Hui Xiong*
NeurIPS'26 Poster: *The Fortieth Annual Conference on Neural Information Processing Systems*

[18] **Less is More: Token-Efficient Video-QA via Adaptive Frame-Pruning and Semantic Graph Integration**
Shaoguang Wang, Jianxiang He, Weiyu Guo, Ziyang Chen, **Yijie Xu**, Xuming Hu, Hui Xiong*
***Findings* of CVPR'26**: *The IEEE / CVF Computer Vision and Pattern Recognition Conference*

[19] **Beyond Boundaries: Leveraging Vision Foundation Models for Source-Free Object Detection**
Huizai Yao, Sicheng Zhao, Pengteng Li, Yi Cui, Shuo Lu, Weiyu Guo, Yunfan Lu, **Yijie Xu**, Hui Xiong*
AAAI'26: *The Fortieth AAAI Conference on Artificial Intelligence*

[20] **See&Trek: Training-Free Spatial Prompting for Multimodal Large Language Model**
Pengteng Li, Pinhao Song, Wuyang Li, Huizai Yao, Weiyu Guo, **Yijie Xu**, Dugang Liu, Hui Xiong*
NeurIPS'25: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*

[21] **Logic-in-Frames: Dynamic Keyframe Search via Visual Semantic-Logical Verification for Long Video Understanding**
Weiyu Guo, Ziyang Chen, Shaoguang Wang, Jianxiang He, **Yijie Xu**, Jinhui Ye, Ying Sun*, Hui Xiong*
NeurIPS'25: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*

[22] **Revisiting Noise Resilience Strategies in Gesture Recognition: Short-Term Enhancement in sEMG Analysis**
Weiyu Guo, Ziyue Qiao, Ying Sun*, **Yijie Xu**, Hui Xiong*
ICML'25: *The Forty-Second International Conference on Machine Learning*

[23] **SpGesture: Source-Free Domain-adaptive sEMG-based Gesture Recognition with Jaccard Attentive Spiking Neural Network**
Weiyu Guo, Ying Sun*, **Yijie Xu**, Ziyue Qiao, Yongkui Yang, Hui Xiong*
NeurIPS'24: *The Thirty-eighth Annual Conference on Neural Information Processing Systems*

SELECTED PROJECTS

Construction Supervision: Regulation-Grounded LLM/VLM System *Sep. 2025 - Present / Poly Group*

- **Screened** and structured evidence from ~17k legal and regulatory documents for construction supervision.
- **Built** a full-stack RAG/VLM system for regulation-grounded assessment and visual construction-site hazard identification.

Multimodal HCI Perception Modeling *Jan. 2023 - Aug. 2024 / MetaHCI*

- **Led** platform setup, data collection, algorithm design, and group coordination for an RBM multimodal HCI project.
- **Developed** LLM-for-AIGC workflows, including training, visualization, deployment, and service maintenance for MetaHCI, a HKUST(GZ)-affiliated startup valued at 70 million RMB.

Political Ideology Analysis on Social Media *Nov. 2022 - Oct. 2023*

- **Built** a 230k+ report dataset from millions of tweets and author profiles.
- **Proposed** a NLP + Temporal Graph Neural Network framework for ideology detection and prediction.

Big Data Mining Services for CRM Systems *Nov. 2022 - May. 2023 / Tencent*

- **Built** web-crawling and pre-trained LM pipelines for large-scale academic talent profiling.
- **Operationalized** extracted profiles for an industry-facing CRM project at Tencent.

INTERNSHIP

Tencent **Qingyun Plan, Yuanbao Search** May 2026 – Present
Research Intern

- **Focuses** on generative retrieval for Yuanbao Search.

SELECTED SCHOLARSHIPS AND AWARDS

Research Travel Grant *Apr. 2025*
Doctoral Scholarship Award *Sep. 2024*
Postgraduate Scholarship Award *Sep. 2022*
First Prize of the *Smart Audiovisual Track*, National AI Application Innovation Challenge *Sep. 2024*
Champion of the *HCI channel*, the 2nd Guangzhou Pazhou Algorithm Competition (**Top 15/3307**) *Sep. 2023*
Scholarship from Nanjing University of Information Science and Technology *Nov. 2021*
Scholarship from China Mobile Communications Corporation (CMCC) *Nov. 2020*
Scholarship from China Mobile Communications Corporation (CMCC) *Nov. 2019*

TEACHING EXPERIENCE

HKUST(GZ)
Big Data Applications for Business and Government (*Spring 2025*)
Natural Language Processing and Its Applications (*Fall 2024*)
Red Bird Challenge Camp (*Summer 2022, 2023, 2024*)

NUIST & Reading
Professional Skills for Mathematicians (*Fall 2021*)
Numerical Analysis (*Fall 2021*)
Applied Stochastic Process (*Fall 2021*)
Mathematical Programming (*Spring 2021*)

SERVICES & SKILLS

PC	**2026**, COLM, ICML, CVPR, ICLR, AAAI, ACM MM **2025**, WWW, AAAI, ACM MM, KDD, NeurIPS, ICML, IJCAI, ICLR, ICASSP, AISTATS, ICWSM **2024**, NeurIPS
Review	ACM Computing Surveys; IEEE Transactions on Circuits and Systems for Video Technology; Transactions on Machine Learning Research (TMLR)
Campus	**Student Representative**, Committee on Undergraduate Studies, HKUST(GZ) Senate (2024–2025)
Stack	Python; PyTorch; Transformers; LLaMA Factory; vLLM & SGLang; LLM MCP&Agents; Web Crawling
Languages	Chinese; English (IELTS 7)